

Video Store Application

Python Programming
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Description Of Application

This programme that is designed to manage a video store's rental system that will enable a the store to to keep record of all their customers as well as their videos and video hires. This system has a server and client side applications. The server application connects to a MySQL database called video_store.sql that store data on customers, videos and video hires and returns . The client side application allows the user to insert, update and request data from the MySQL video_store.sql database.

This Programme has four Functions:

1. Registering Customers
2. Registering Videos
3. Hiring Videos
4. Returning Videos

System Requirements

To use the Video Store application the PC you will be running the program on must have the following installed:

1. Python
2. mySQL Workbench
3. Any Python Friendly IDE such as: Visual Studio Code, PyCharm Community or Python IDLE .

Installation and Setup

1. Install Python
 - a. Visit this link (<https://www.python.org/downloads/>) and download the latest version of Python. Once downloaded, Double Click to Install on PC.
2. Install one of the following Python friendly IDE's
 - a. Python IDLE is automatically installed when you install Python.
 - b. To install Visual Studio Code visit this link <https://code.visualstudio.com/> and download the latest version of Visual Studio Code. Once downloaded, double click to Install on PC
 - c. To install PyCharm Community visit this link (<https://www.jetbrains.com/pycharm/download/?section=windows>) and download the latest version of PyCharm Community. Once downloaded, double click to Install on PC
3. To install mySQL Workbench download mySQL Installer (<https://dev.mysql.com/downloads/installer/>) . Double Click to Install on PC. When setting up you Workbench environment, ensure you install a Python Connector. After installation, set up connection with username and password.
4. Once all the above have been installed Open the connection and select File on top left of screen, then select Open File. Navigate to the Video Store File and double click on the video_store.sql file. This will open the script that will create the database .Click the yellow lighting arrow to run the script and create the database. Then open the Video Store folder and double click on the Video Story.py file and select your chosen IDE. Once the program source code is open, go to line ten of the source code and change the user, password and host values to the ones used on the mySQLWorkbench Connection you just ran the video_store.msql script with. After this, You can run the Application on the IDE.

Directions On How To Use The Application

This program has four functions:

1. Registering Customers

- To register customer select option 1 the the appilacrction will prompt you to enter the customers phone number. The application will check if the customer's phone number already exists in the system. If the customer's phone number is already in the system database the the application will inform user that the customer already exists. If the customer phone number is not in the database, then the application will prompt the user to enter the customer's details (name, surname, address) . The application will display that the customer has successfully registered and it will return to the main page

2. Registering Videos

- To register videos to the system select options 2 and the application will prompt you to enter the movie name and type video type (R for a new video or B for an old video). The application will display that the customer has successfully registered and it will return to the main page

3. Hiring Videos

- To hire a video out to a customer select option 3 and the application will prompt the user to enter the customer phone number. The system will check if the customer exists in the database. If the customer doe not exist the application will inform the user and return to the main page to allow the user to register the customer. If the customer does exist the program will prompt the user to enter the video ID of the video being hired. The system will check if then the video being hired exists in the system. if the video doesn't exist in the system will inform the user that it does not exist and return to the main page. if the video does exist in the system, the application will insert a record of the video hire into the database with the customer details and current date.

4. Returning Videos

- To record a video being returned by a customer the user will be prompted to enter the video id of video being returned. the system will check the database for a record of the video hire. If the hire record is found it will be updated with the return date (current date).

Screenshots Of The Application

```
=====
|                               VIDEO STORE                               |
=====
| 1. Register Customer                                                  |
| 2. Register Video                                                    |
=====
| 3. Hire Out Video                                                    |
| 4. Return Video                                                      |
=====
| X. Exit                                                              |
=====
Choice:_1
Enter Customer's 10 digit Phone Number: 0723514864
Customer Name: Angel
Customer Surname: Siwele
Customer Home Address: Nelspruit
Customer Successfully Registered
```

```
=====
|                               VIDEO STORE                               |
=====
| 1. Register Customer                                                  |
| 2. Register Video                                                    |
=====
| 3. Hire Out Video                                                    |
| 4. Return Video                                                      |
=====
| X. Exit                                                              |
=====
Choice:_2
Please Enter video name: Barbie
Please Enter video type (R for New video or B for old videos): r
Video Successfully Registered
```

```
=====
|                               VIDEO STORE                               |
=====
| 1. Register Customer |
| 2. Register Video   |
=====
| 3. Hire Out Video   |
| 4. Return Video     |
=====
| X. Exit |
=====
Choice:_3
Enter Customer's 10 digit Phone Number: 0723514864
Enter Video ID 1
Video Hire Successfully Recorded
```

```
=====
|                               VIDEO STORE                               |
=====
| 1. Register Customer |
| 2. Register Video   |
=====
| 3. Hire Out Video   |
| 4. Return Video     |
=====
| X. Exit |
=====
Choice:_4
Please enter video id of video being returned 1
Enter Customer's 10 digit Phone Number: 0723514864
Video Return Successfully Recorded
```

Project Source Code

```
#CURDATE Error: Change customer_table_query to match workbech customer
table. Issue was hired and videoVer
import datetime
import mysql.connector
conn = mysql.connector.connect(user="root", password="root",
host="localhost", database="video_store")
cur = conn.cursor()

current_date= datetime.datetime.now()
#print(str(current_date)[0:10])
'''
#create database if not exit
database_query = "CREATE DATABASE IF NOT EXISTS  video_store;"
cur.execute(database_query)
conn.commit()
#create tables if not exit
customer_table_query = "\
CREATE TABLE IF NOT EXISTS customers(\
custId INT PRIMARY KEY AUTO_INCREMENT,\
fname VARCHAR(40)  NOT NULL,\
sname VARCHAR(40)  NOT NULL,\
address VARCHAR(40)  NOT NULL,\
phone VARCHAR(10) NOT NULL UNIQUE  \
);"
cur.execute(customer_table_query)
conn.commit()

#create tables if not exit
video_table_query = "\
CREATE TABLE IF NOT EXISTS videos(\
VideoId INT PRIMARY KEY AUTO_INCREMENT,\
videoVer INT NOT NULL ,\
vname VARCHAR(15) NOT NULL,\
video_type VARCHAR(1) NOT NULL ,\
dateAdded VARCHAR NOT NULL \
```

```

);"
cur.execute(video_table_query)
conn.commit()

#create tables if not exist
hire_table_query = "\
CREATE TABLE IF NOT EXISTS hire(\
custId INT NOT NULL,\
videoId INT NOT NULL,\
dateHired VARCHAR(20) NOT NULL,\
dateReturn VARCHAR(20),\
FOREIGN KEY (videoId) REFERENCES videos(videoId),\
FOREIGN KEY (custId) REFERENCES customers(custId))\
;"
cur.execute(hire_table_query)
conn.commit()
'''
menu = "===== " + "\n"
menu = menu + "|                VIDEO STORE                |"
+ "\n"
menu = menu + "===== "
+ "\n"
menu = menu + "| 1. Register Customer                        |"
+ "\n"
menu = menu + "| 2. Register Movie                          |"
+ "\n"
menu = menu + "===== "
+ "\n"
menu = menu + "| 3. Hire Out Movie                          |"
+ "\n"
menu = menu + "| 4. Return Movie                            |"
+ "\n"
menu = menu + "===== "
+ "\n"
menu = menu + "| X. Exit                                    |"
+ "\n"
menu = menu + "===== "
+ "\n"
menu = menu + "Choice:_"

```

```

def main():

    possibleAnswers = ["1", "2", "3", "4", "x", "X"]
    answer = input(menu)
    while answer not in possibleAnswers:
        print("Please enter valid option")
        main()
    else:
        #Exit
        if answer == "x" or answer == "X":
            exit()
        #Register Customer
        elif int(answer) == 1:
            phonenumber = input("Enter Customer's 10 digit Phone Number: ")

            while len(phonenumber) != 10:
                phonenumber = input("Enter Customer's 10 digit Phone Number: ")

            else:
                while search_phone(phonenumber) == True:
                    print("Customer already in database")
                    main()
                else:
                    name = input("Customer Name: ")
                    surname = input("Customer Surname: ")
                    homeaddress = input("Customer Home Address: ")
                    # insert details into customers table
                    query = "INSERT INTO customers (fname,sname,address,phone) VALUES ('"+ name + "', '"+ surname + "', '"+ homeaddress + "', '"+ phonenumber + "')"
                    cur.execute(query)
                    conn.commit()
                    print("Customer Successfully Registered")
                    main()

        #2 Register Movie
        elif int(answer) == 2:
            moviename = input("Please Enter movie name: ")

```

```

        #if vName already exists, set new video Id to existing videoId
        movietype = input("Please Enter movie type (R or B): ")
        while movietype != "R" and movietype != "r" and movietype !=
"B" and movietype != "b":
            movietype = input("Please Enter Valid Movie Type (R or B):
")

        if search_video(moviename) == True:
            movieId = int(get_videoId(moviename))
            query =f'INSERT INTO videos
(videoId,vname,video_type,dateAdded) VALUES
({movieId},{moviename},{movietype},{SELECT CURDATE()}'
            cur.execute(query)
            conn.commit()
            print("Video Successfully Registered")
            main()
        else:
            query =f'INSERT INTO videos (vname,video_type,dateAdded)
VALUES ({moviename},{movietype},CURDATE()) '
            cur.execute(query)
            conn.commit()
            print("Video Successfully Registered")
            main()

#Hire Out Movie
elif int(answer) == 3:
    phonenumber = input("Enter Customer's 10 digit Phone Number:
")

    while len(phonenumber) != 10:
        phonenumber = input("Enter Customer's 10 digit Phone
Number: ")

    else:
        if search_phone(phonenumber) == False:
            print("Customer not Registered")
            main()
        else:
            videoId = input("Enter Video ID ")

```

```

        query = f'INSERT INTO hire (custId,videoId,dateHired)
VALUES ((SELECT custId FROM customers WHERE phone =
"{phonenummer}"),{int(videoId)}, (SELECT CURDATE())) '
        cur.execute(query)
        conn.commit()
        print("Video Hire Successfully Recorded")
        main()

#Return Video
elif int(answer) == 4:
    videoId = input("Please enter video id of movie being returned
")
    phonenummer = input("Enter Customer's 10 digit Phone Number:
")
    get_custId_query = f'SELECT custId FROM customers WHERE phone
= {phonenummer}'
    cur.execute(get_custId_query)
    conn.commit()
    customerId = cur.fetchall()[0][0]

    if get_hire_record(customerId,videoId) == True:
        query = f'UPDATE hire SET dateReturn = (SELECT CURDATE())
WHERE videoId = {videoId} AND custId = (SELECT custId FROM customers WHERE
phone = {phonenummer}) '
        cur.execute(query)
        conn.commit()
        print("Video Return Successfully Recorded")
        main()
    else:
        print("There is no record of this video being hired by
this customer")
        main()

def search_phone(number):
    query = f'SELECT * FROM customers WHERE phone = {number}'
    cur.execute(query)
    result = cur.fetchall()
    return len(result)>0

```

```
def search_video(video_name):
    query = f'SELECT * FROM videos WHERE vname = {video_name}'
    cur.execute(query)
    result = cur.fetchall()
    return len(result)>0

def get_videoId(video_name):
    query = f'SELECT * FROM videos WHERE vname = {video_name}'
    cur.execute(query)
    result = cur.fetchall()
    #print(result)
    return result[0][0]

def get_hire_record(customer_id,video_id):
    query = f'SELECT * FROM hire WHERE vname = {video_id} AND custId = {customer_id}'
    cur.execute(query)
    result = cur.fetchall()
    return len(result)>0

main()
conn.close()
```

Program Design

Video Store System Flow Chart

